

Assignment 3: The Discrete Geometric Conservation Law

- The assignment should be handed in at the latest Wednesday 25th March 2026 12:00 in order to receive full credit, otherwise a 2 point penalty is applied to the grade of the assignment.
- The assignment can be made in groups of **at most 3 students**.
- Please be sure to include the names and student numbers in your report.
- Please hand in an electronic version (also a scanned version of a (clearly) hand-written report is accepted) through Brightspace (only one submission per group is required).
- You do not have to make a full report (including summary, list of figures, list of symbols etc), but indicate clearly the question numbers when giving your answer / results / discussion of results.
- When an implementation is requested, please provide (only) the code snippets where you made your implementation.

Introduction

In fluid-structure interaction simulations, the fluid equations are solved on a deforming mesh. An often used method for dealing with deforming meshes is the Arbitrary Lagrangian-Eulerian (ALE) approach, which allows the fluid dynamics equations to be solved on arbitrarily moving/deforming control volumes. When using the ALE approach, a Geometric Conservation Law (GCL) can be identified that ties the mesh deformation to mesh velocities, used in the governing equations. In this exercise we will investigate the GCL for a range of simple test problems.

Governing equations

A general formulation for the fluid dynamics equations on a domain Ω is

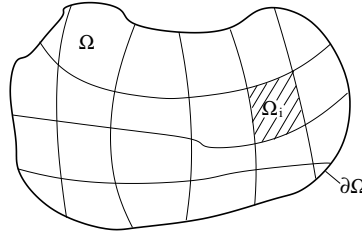


Figure 1: Computational domain

$$\frac{d}{dt} \int_{\Omega} \mathbf{U} dV + \oint_{\partial\Omega} \left[\mathbf{F}(\mathbf{U}) - \mathbf{U} \frac{d\vec{x}}{dt} \right] \cdot \vec{n} dS = 0, \quad (1)$$

wherein $\mathbf{U}$ the conservative variables $(\rho, \rho\vec{u}, \rho E)$, $\mathbf{F}$ the flux vector containing convective and diffusive terms, $\frac{d\vec{x}}{dt}$ the mesh velocity, $\vec{n}$ the outward pointing normal to surface $\partial\Omega$. Suppose we discretize the domain Ω using a finite number of volumes Ω_i , we can write for a single volume

$$\frac{d}{dt} \int_{\Omega_i} \mathbf{U} dV + \oint_{\partial\Omega_i} \left[\mathbf{F}(\mathbf{U}) - \mathbf{U} \frac{d\vec{x}}{dt} \right] \cdot \vec{n} dS = 0. \quad (2)$$

This will be the starting point for deriving the continuous Geometric Conservation Law, and the semi-discrete and discrete versions.

1 Geometric Conservation Law

One of the viewpoints on the significance of the GCL is that it ensures uniform flow to remain uniform on arbitrarily moving meshes. This feature is used to derive first of all the continuous GCL.

Continuous GCL

In the derivation we assume $\mathbf{U} = \mathbf{U}_0$ to be uniform in the domain Ω and hence one may expect that $\mathbf{F}_0 = \mathbf{F}(\mathbf{U}_0)$ also uniform. The control volume Ω_i has a closed surface $\partial\Omega_i$.

Exercise 1.1 Derive the GCL by substitution of $\mathbf{U}_0$ into (2)

Note that up till now, the relation still uses continuous functions and the continuous GCL is independent of the discretization method used for numerically solving the fluid dynamics equations. For certain methods, such as space-time discretizations, the GCL is generally satisfied in a natural way. For other discretization approaches, such as finite-volume discretizations, this may not be the case. Since a major part of engineering CFD codes use the finite volume approach, we continue in a finite-volume framework for obtaining the semi-discrete formulation of the GCL.

Semi-discrete GCL

First the control volume Ω_i is discretized to V_i and its boundary $\partial\Omega_i$ to a summation of discrete faces (or edges in 2D), as shown in Fig. 2. The spatial integrals are discretized using

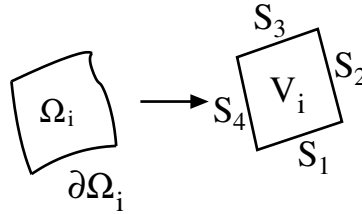


Figure 2: Discrete domain of a single control volume V_i .

$$\oint_{\partial\Omega_i} \frac{d\vec{x}}{dt} \cdot \vec{n} dS \approx \sum_{faces} \left(\frac{d\vec{x}}{dt} \cdot \vec{n} S \right)_{face}, \quad (3)$$

with S the face surface area for each discrete face, so that we obtain the semi-discrete GCL

$$\frac{d}{dt} V_i = \sum_{faces} \left(\frac{d\vec{x}}{dt} \cdot \vec{n} S \right)_{face}. \quad (4)$$

Exercise 1.2 What do the left and right sides of (4) represent (what is their physical interpretation)?

Discrete GCL

In (4) we still have a temporal derivative. To obtain a fully discrete equation a wide variety of time integration schemes can be chosen to discretize the temporal derivative. Here we will consider four implicit integration schemes:

- **Backward Euler** The first order Backward Euler scheme is quite popular due to its simplicity and stability. The integration of a quantity ϕ yields

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} + R(\phi^{n+1}) = 0, \quad (5)$$

wherein $R(\phi^{n+1})$ is the spatial discretization, evaluated at the implicit time level t_{n+1} .

- **BDF2** The second order Backward Differencing scheme is often used in CFD codes. It obtains a second order accuracy by using the state of an additional time level t_{n-1} . The integration of a quantity ϕ yields

$$\frac{\alpha_1 \phi^{n+1} + \alpha_2 \phi^n + \alpha_3 \phi^{n-1}}{\Delta t} + R(\phi^{n+1}) = 0, \quad (6)$$

wherein $\vec{\alpha} = (\frac{3}{2}, -2, \frac{1}{2})$ for BDF2 and $R(\phi^{n+1})$ is the spatial discretization, evaluated at the implicit time level t_{n+1} . Since the method needs data at time level t_{n-1} , which is generally unavailable at $t_n = t_0$, the first time step is generally performed with a Backward Euler scheme for which $\vec{\alpha} = (1, -1, 0)$.

As shown in the lecture, the change in volume of the cell is decomposed into volume changes caused by the motion of each cell face individually:

$$V^{n+1} - V^n = \sum_{faces} \Delta V_{face}^{n+1}, \quad (7)$$

wherein ΔV_{face}^{n+1} the so-called swept-volume for a face moving from its location at t_n to its location at t_{n+1} . Therefore, we obtain for the Backward Euler scheme the following Discrete-GCL (D-GCL)

$$\frac{V^{n+1} - V^n}{\Delta t} = \sum_{faces} \frac{\Delta V_{face}^{n+1}}{\Delta t} = \sum_{faces} \left(\frac{d\vec{x}}{dt} \cdot \vec{n} S \right)_{face}^{n+1}, \quad (8)$$

which is enforced for each face individually

$$\frac{\Delta V_{face}^{n+1}}{\Delta t} = \left(\frac{d\vec{x}}{dt} \cdot \vec{n} S \right)_{face}^{n+1}, \quad (9)$$

so that the face normal velocity $(\frac{d\vec{x}}{dt} \cdot \vec{n})^{n+1}$ should be

$$\left(\frac{d\vec{x}}{dt} \cdot \vec{n} \right)^{n+1} = \frac{\Delta V_{face}^{n+1}}{\Delta t S}, \quad (10)$$

in order to ensure that the D-GCL is satisfied.

Exercise 1.3 Derive the D-GCL for the BDF2 scheme

2 Simulation of linear advecting flow on a one-dimensional moving mesh

Next we investigate the simulation of linear advecting flow on a moving mesh and the disturbances that can be generated when not satisfying the D-GCL. The matlab files necessary

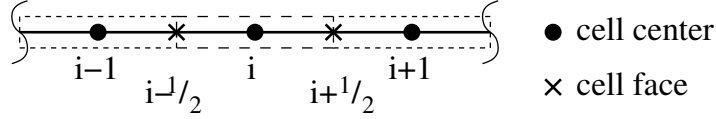


Figure 3: One dimensional mesh.

are stored in folder `matlab\DGCL-1D\`. We start with the example shown in the lectures for a one-dimensional testcase (Fig. 3) for which an exact mesh motion is known:

$$x = x_0 + A_m \sin(2\pi x_0) \sin(2\pi t), \quad (11)$$

$$\frac{dx}{dt} = 2\pi A_m \sin(2\pi x_0) \cos(2\pi t), \quad (12)$$

wherein $x_0 = x(t = 0)$ and A_m is the amplitude of the mesh motion. For the initial mesh at $t = 0$ the domain is uniformly divided into N finite volume cells to discretize the domain $[0,1]$. The model that is discretized is given in (2). We assume a constant advection speed u_a and we will solve for the fluid density ρ , using only the continuity equation, i.e.

$$\frac{d}{dt} \int_{\Omega_i} \rho dV + \oint_{\partial\Omega_i} \left[\rho \left(u_a - \frac{dx}{dt} \right) \right] \cdot n dS = 0, \quad (13)$$

for which the boundary integral reduces to a summation of a left and right face contribution:

$$\frac{d}{dt} (\rho V)_i + \left[\rho \left(u_a - \frac{dx}{dt} \right) \right]_{i+1/2} - \left[\rho \left(u_a - \frac{dx}{dt} \right) \right]_{i-1/2} = 0, \quad (14)$$

wherein we use a simple average to approximate the density at the cell faces $i \pm 1/2$:

$$\rho_{i+1/2} = \frac{\rho_i + \rho_{i+1}}{2}. \quad (15)$$

When e.g. a Backward Euler time integration is chosen, (14) and (15) result in

$$\frac{\rho_i^{n+1} V_i^{n+1} - \rho_i^n V_i^n}{\Delta t} + \frac{\rho_i^{n+1} + \rho_{i+1}^{n+1}}{2} \left(u_a - \frac{dx}{dt} \right)_{i+1/2}^{n+1} - \frac{\rho_{i-1}^{n+1} + \rho_i^{n+1}}{2} \left(u_a - \frac{dx}{dt} \right)_{i-1/2}^{n+1} = 0, \quad (16)$$

or,

$$\begin{aligned} & \left[V_i^{n+1} + \frac{\Delta t}{2} \left(u_a - \frac{dx}{dt} \right)_{i+1/2}^{n+1} - \frac{\Delta t}{2} \left(u_a - \frac{dx}{dt} \right)_{i-1/2}^{n+1} \right] \rho_i^{n+1} \\ & + \left[\frac{\Delta t}{2} \left(u_a - \frac{dx}{dt} \right)_{i+1/2}^{n+1} \right] \rho_{i+1}^{n+1} - \left[\frac{\Delta t}{2} \left(u_a - \frac{dx}{dt} \right)_{i-1/2}^{n+1} \right] \rho_{i-1}^{n+1} = V_i^n \rho_i^n, \end{aligned} \quad (17)$$

which results in a system

$$L \boldsymbol{\rho}^{n+1} = R \boldsymbol{\rho}^n. \quad (18)$$

The initial condition for the density field is a uniform density of 1 with a sin-shape perturbation, for which the amplitude **Amp** can be chosen:

$$\rho(x, 0) = 1 + \text{Amp} \sin(2\pi x). \quad (19)$$

For the linear advection case with periodic boundary conditions, the exact solution is given by

$$\rho(x, t) = 1 + \text{Amp} \sin(2\pi(x - u_a t)). \quad (20)$$

The matlab program `dgcl_BE.m` sets-up this system of equations and solves the system for every time step. The influence of the DGCL is through the $(\frac{dx}{dt})$ terms in L .

Backward Euler For the Backward Euler scheme, the swept volumes of the faces reduce to a change in position of the face coordinate χ and therefore its D-GCL (10) becomes

$$\left(\frac{dx}{dt}\right)^{n+1} = \frac{\chi^{n+1} - \chi^n}{\Delta t}, \quad (21)$$

which is implemented into the matlab file `dgcl_BE.m`, along with the options to choose the exact mesh velocities at $t_{n+1/2}$ and t_{n+1} .

Exercise 2.1 In `dgcl_BE.m` we set the number of finite volumes $N = 20$, the final simulation time `tend = 2`, the amplitude of the mesh motion `A_m = 0.1` and for each `method=1,2,3` we set the time step and plot format according to Table 1. We consider only a uniform solution for now `Amp = 0`, and we do not include the linear advection `u.a = 0`. This way the only errors we encounter can be caused by the contribution of the mesh motion. When switching from one method to the other, you can close the figures to start a clean plot. You can verify that satisfying the D-GCL preserves the constant solution. As a rough estimate for the order with which the error reduces with respect to Δt , one can judge the amplitude decrease in the solution at t_{end} for various Δt , e.g. when the amplitude $\approx$ halves when Δt is halved, the error reduces with $O(\Delta t)$. Which orders do you find approximately for `method=1,2`?

Table 1: Time step and format settings.

case	Δt	fmt
1	0.1	'-b'
2	0.05	'-r'
3	0.025	'-g'

We are going to make a more thorough analysis of the error introduced by not satisfying the D-GCL by considering two errors:

- The root-mean-square error at the final time,
- The root-mean-square error over the whole simulation.

The first error is defined by

$$\epsilon_1 = \sqrt{\frac{\sum_{i=1}^N (\rho_i^M - \rho_{ex}(x_i, t_M))^2}{N}}, \quad (22)$$

wherein ρ_{ex} the exact solution, N the total number of unknowns in space and M the total number of time steps taken. The second error is defined by

$$\epsilon_2 = \sqrt{\frac{\sum_{j=1}^M \sum_{i=1}^N (\rho_i^j - \rho_{ex}(x_i, t_j))^2}{NM}}. \quad (23)$$

We perform a time step refinement study by running the simulation at time steps $\Delta t = 0.1 \cdot 2^{-k}$, $k = 0, 1, 2, 3, 4, 5, 6$. Note that by setting `show_sol = 0` in the Matlab script, one can speed up the computation since the intermediate results are not shown. For each of the chosen time steps, determine the errors ϵ_1 and ϵ_2 for `method=1,2,3`. By visualizing the ϵ versus Δt results in a log-log plot, one can determine the order with which the error decreases as Δt decreases by measuring the slope of the curve in the plot. When the time steps are small enough and the method is consistent, the error should behave like

$$\epsilon \approx c \Delta t^p, \quad (24)$$

when the error decreases with $O(\Delta t^p)$. Therefore, taking the logarithm on both sides yields

$$\log \epsilon \approx \log c + p \log \Delta t, \quad (25)$$

which in a log-log plot is similar to a linear function $Y = C + pX$ (with $Y = \log \epsilon$ and $X = \log \Delta t$), with a slope of p .

Exercise 2.2 In `dgcl.BE.m` implement the computation of the errors ϵ_1 and ϵ_2 ; set `N = 20`, `tend = 2`, `A_m = 0.1`, `Amp = 0`, `u_a = 0` and for each `method=1,2,3` determine the error for the time steps $\Delta t = 0.1 \cdot 2^{-k}$, $k = 0, 1, 2, 3, 4, 5, 6$ and visualize the results in a log-log plot. From the plot determine and report the order of consistency with which the errors tend to zero when the time step is reduced. Even though the error should be zero when satisfying the D-GCL one may still find an error unequal to zero - what kind of error is observed here?

As a second step we will analyse the error by making a Taylor series expansion around the location in time for which the exact velocity is prescribed in `method = 1,2`; as you know the Taylor series expansion about t_n for a small perturbation δ yields

$$x(t_n + \delta) = x(t_n) + \delta \left. \frac{dx}{dt} \right|_{t_n} + \frac{\delta^2}{2} \left. \frac{d^2x}{dt^2} \right|_{t_n} + \frac{\delta^3}{6} \left. \frac{d^3x}{dt^3} \right|_{t_n} + \frac{\delta^4}{24} \left. \frac{d^4x}{dt^4} \right|_{t_n} + O(\delta^5), \quad (26)$$

for a polynomial approximation of degree 4. Notice that the time derivatives of x are taken at the time instance t_n . We know how to satisfy the D-GCL for the Backward Euler time integration scheme: use Eq. (21). As the coordinated χ of the faces are in fact spatial locations in x , we can also write, for a given face location at x , that satisfying the D-GCL yields:

$$\left(\frac{dx}{dt} \right)^{n+1} = \frac{x^{n+1} - x^n}{\Delta t}. \quad (27)$$

Therefore when using the exact mesh velocity at t_{n+1} , instead of the D-GCL, the difference between the two (the error) is

$$\epsilon = \frac{x^{n+1} - x^n}{\Delta t} - \left. \frac{dx}{dt} \right|_{t_{n+1}}. \quad (28)$$

We can now use a Taylor expansion around t_{n+1} to express x^n as

$$x^n = x^{n+1} - \Delta t \left. \frac{dx}{dt} \right|_{t_{n+1}} + \frac{\Delta t^2}{2} \left. \frac{d^2x}{dt^2} \right|_{t_{n+1}} - \frac{\Delta t^3}{6} \left. \frac{d^3x}{dt^3} \right|_{t_{n+1}} + \frac{\Delta t^4}{24} \left. \frac{d^4x}{dt^4} \right|_{t_{n+1}} + O(\Delta t^5), \quad (29)$$

which, when substituted into (28) provides the approximation of the error

$$\epsilon = \left[\left. \frac{dx}{dt} \right|_{t_{n+1}} - \left. \frac{dx}{dt} \right|_{t_{n+1}} \right] - \frac{\Delta t}{2} \left. \frac{d^2x}{dt^2} \right|_{t_{n+1}} + O(\Delta t^2), \quad (30)$$

which shows that the leading error when using the exact mesh velocity at t_{n+1} instead of the D-GCL is only first order accurate with Δt .

Exercise 2.3 Write the Taylor series expansions up to degree 4 about $t_{n+1/2}$ for x^n and x^{n+1} and determine the leading error when using the exact velocity at $t_{n+1/2}$. Do the numerical results obtained in Exercise 2.2 correspond to the analytical results? Which error measure is more precise: ϵ_1 (determining the error only based on the final result) or ϵ_2 (determining the error based on all time steps during the simulation)?

Next we perform a similar analysis on the BDF2 scheme.

Exercise 2.4 Write the D-GCL you derived for the BDF2 scheme in terms of the face locations χ^{n-1} , χ^n , χ^{n+1} and the coefficients $\vec{\alpha}$.

Implement the D-GCL into `dgcl_BDF2.m`. Verify you implemented the D-GCL correctly so that the constant solution is preserved.

Perform and present a similar analysis as was done in Exercises 2.2, to determine graphically the order of consistency.

Solving flow on a deforming domain

As a final investigation, we check the performance of solving linear advecting flow on a deforming domain using the BDF2 scheme. In this case we solve for the advection in combination with the mesh motion. Also, we impose an initial solution that is not uniform by setting `Amp = 0.1`. To improve the spatial resolution, the mesh is refined and a time step refinement study is done.

Exercise 2.5 We set `N = 200`, `tend = 2`, `A_m = 0.1`, `Amp = 0.1`, `u_a = 1` and for each `method=1,2,3` determine the error for the time steps $\Delta t = 0.1 \cdot 2^{-k}$, $k = 0, 1, 2, 3, 4, 5, 6$ and visualize the results in a log-log plot.

Discuss the results. When satisfying the D-GCL, should we now obtain the exact solution? Is satisfying the D-GCL still important when in reality we typically do not solve for uniform flow?